



KEVIN EKA PRATAMA

BACKEND & CLOUD SYSTEMS ENGINEER

Distributed Microservices • High-Concurrency Go & Java Spring Boot • ACID Ledgers

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EXECUTIVE SUMMARY

TARGET: BACKEND (PAGE 1 OF 3)

Backend & Cloud Systems Engineer with 2+ years of enterprise Application Support experience at PT PLN Icon+. Proven track record maintaining 100% SLA compliance for production operational tickets, authoring complex SQL queries (PostgreSQL, Oracle, MySQL) for data auditing, and monitoring 24/7 high-availability system workflows. Deeply proficient in AI-Assisted Engineering, pairing with LLM tools to accelerate schema design, boilerplate generation, and integration testing. Independently architected and deployed 19 production-grade backend microservices and cloud systems using Go (Golang), Java Spring Boot 3.3, Bun/Hono, and Express.js. Strong mastery in Clean Architecture (DDD), ACID transactional ledgers with row-level locks, Redis cache-aside patterns, RabbitMQ message brokers, gRPC binary protocols, and Docker containerization.

PROFESSIONAL EXPERIENCE

Application Support Engineer • PT PLN Icon+

2023 - Present

- Investigated and resolved technical operational and incident tickets for enterprise utility platforms with a strict **100% SLA compliance rate**, ensuring zero-downtime operations.
- Authored and executed complex SQL queries across **PostgreSQL, Oracle, and MySQL** for transactional data validation, data auditing, anomaly rectification, and executive operational reporting.
- Monitored nationwide enterprise system workflows and microservices 24/7, proactively identifying transaction failures, API synchronization bottlenecks, and external service latency spikes.
- Investigated application log telemetry (trace IDs, stack traces), conducted Root Cause Analyses (RCA), and collaborated directly with core backend developers to verify and deploy production bug hotfixes.
- Documented recurring operational incident runbooks and standardized SQL query templates, accelerating triage duration for similar tickets and staging UAT patch verifications.

AI-Assisted Backend Systems & Architecture • Independent Engineering & Open Source Projects

2023 - Present

- Pioneered human-AI pair programming workflows with frontier models (**Gemini 2.5, Claude 3.7, OpenAI APIs**) to accelerate architectural design, relational schema modeling, and automated test suite generation.
- Independently architected, developed, and audited **19 production-grade backend microservices** applying Clean Architecture (DDD) and Docker containerization.
- Architected normalized relational database schemas with structured foreign keys, strict server-side schema validation, and decoupled domain layers (handler, service, repository) for long-term maintainability.
- Developed a digital wallet simulation lab exploring concurrency safety and preventing double-spending via **PostgreSQL row-level locks (SELECT FOR UPDATE)** and Redis caching.
- Audited microservice security contracts, implemented Token Bucket distributed rate limiting, and maintained containerized orchestration environments.

TECHNICAL COMPETENCIES & CORE STACK

Backend Languages: Go (Golang 1.25/1.26), Java (JDK 17/21), TypeScript, JavaScript (Node.js/Bun), Python 3, SQL, Bash

Frameworks & Architecture: Java Spring Boot 3.3 (Spring Security 6, JPA), Go (Gin/Fiber/Echo), Bun + Hono, Express.js, FastAPI, Clean Architecture (DDD), gRPC (Protobuf), RESTful APIs, Microservices

Databases & Message Brokers: PostgreSQL (GORM, Prisma, ACID Transactions, Connection Pooling, Row-level Locks), MySQL, MongoDB (NoSQL), Redis (Cache-Aside, Rate Limiting), RabbitMQ (Message Broker)

AI & Agentic Engineering: Gemini 2.5, Claude 3.7, OpenAI APIs, AI Component Prototyping, Prompt Engineering, Agentic Tooling, Automated Testing

DevOps, Cloud & Tooling: Docker, Docker Compose, Linux Bash, Git & GitHub, Postman, Swagger / OpenAPI 3.0, CI/CD GitHub Actions

EDUCATION

Bachelor of Computer Science / Information Technology (S.Kom) • Universitas AMIKOM • GPA: 3.42 / 4.00 (Thesis Defense: Dec

2017 -
2023

2022 | Official Degree / Graduation: 2023)

PILLAR 1: DISTRIBUTED GO & JAVA SYSTEMS

- **go-distributed-microservices-lab** — Distributed microservices architecture with binary gRPC, RabbitMQ message brokers, and Redis cache-aside.
Tech Stack: Go, gRPC, Protobuf, RabbitMQ, Redis, Worker Pools, Docker
- **go-ecommerce-gateway-engine** — High-throughput reverse proxy API gateway with MongoDB, order lifecycle engine, and Swagger OpenAPI.
Tech Stack: Go 1.26, Gin, MongoDB, Reverse Proxy, Swagger OpenAPI
- **go-banking-core-system** — Digital wallet balance transfer simulation exploring ACID transactions, row-level locks, and Bcrypt PIN.
Tech Stack: Go, Echo, PostgreSQL, ACID Row Locks, Bcrypt PIN, Swagger UI
- **go-clean-arch** — Domain-driven Clean Architecture REST API decoupling business usecases from repository persistence.
Tech Stack: Go, Clean Architecture (DDD), Domain/Usecase/Repository, PostgreSQL
- **go-rest-api-enterprise** — Enterprise Go boilerplate featuring Zap structured logging, cache-aside Redis, and graceful shutdown.
Tech Stack: Go, Gin, GORM, Redis Cache-Aside, Zap Structured Logging, Docker
- **spring-boot-enterprise-platform** — Enterprise backend platform with Spring Security JWT, AOP logging, Bucket4j rate limiting, and Docker.
Tech Stack: Java 17, Spring Boot 3.3, Spring Security JWT, Bucket4j Rate Limiter, Docker

PILLAR 2: CLOUD APIS & TYPESCRIPT MICROSERVICES

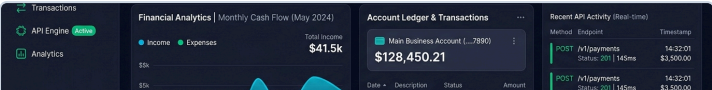
- **hono-ecommerce-engine** — Sub-millisecond REST API engine running on Bun runtime with Drizzle ORM and live WebSocket chat.
Tech Stack: Bun Runtime, Hono v4, Drizzle ORM, WebSocket Live Chat, SQLite
- **express-prisma-realworld-api** — Conduit specification publication backend built with Express, TypeScript, Prisma ORM, and Nx monorepo.
Tech Stack: Express.js, TypeScript, Prisma ORM, Nx Monorepo, JWT, Jest Suite
- **express-typescript-prisma-api** — Type-safe modular REST API built with Express v5, TypeScript, Prisma 7, and LibSQL adapter.
Tech Stack: Express v5, TypeScript, Prisma 7 ORM, LibSQL Adapter, tsx
- **express-prisma-product-api** — Modular product catalog REST API with JWT authentication, Multer uploads, and Zod schema validation.
Tech Stack: Express v5, TypeScript, Prisma ORM, JWT Auth, Multer, Zod Validation
- **express-sqlite-ecommerce-api** — Lightweight e-commerce API with SQLite prepared statements, atomic checkout transactions, and Swagger UI.
Tech Stack: Express v5, SQLite Prepared Statements, ACID Transactions, Swagger UI
- **express-realtime-api-service** — Real-time event broadcasting API engine integrating Socket.IO, dual DB (MongoDB + MySQL), and Winston.
Tech Stack: Express v5, Socket.IO, Dual DB (MongoDB + MySQL), Winston, Zod

PILLAR 3: SPECIALIZED MICROSERVICES, WEBHOOKS & DATA PIPELINES

- **express-prisma-payment-api** — Payment processing backend with Midtrans webhook verification, automated PDFKit invoices, and Nodemailer.
Tech Stack: Express.js, Midtrans Webhook, PDFKit Invoicing, Nodemailer
- **AI-api-manager** — API reverse proxy gateway with API key validation, Token Bucket rate limiting, and quota analytics.
Tech Stack: Node.js, Reverse Proxy Gateway, Redis Rate Limiting, API Key Auth, React UI
- **spring-boot-book-manager-api** — Book catalog microservice with Java 17, Spring Boot 3.3, Spring Data MongoDB, and OpenAPI 3.0.
Tech Stack: Java 17, Spring Boot 3.3, Spring Data MongoDB, OpenAPI 3.0
- **express-book-catalog-api** — Real-time book inventory management API with Express v5, Prisma 7, Socket.IO, and Redis rate limiting.
Tech Stack: Express v5, Prisma 7, Socket.IO Real-time, Redis Rate Limiter, Jest
- **express-redis-url-shortener** — URL redirect engine leveraging Redis cache-aside for sub-millisecond lookups and MongoDB storage.
Tech Stack: Express, Redis Cache-Aside, Sub-millisecond Redirects, MongoDB
- **express-mongo-content-api** — Content management REST API with MongoDB Mongoose, Redis caching, Socket.IO, and cron schedulers.
Tech Stack: Express, MongoDB Mongoose, Redis Caching, Socket.IO, Cron Jobs
- **express-mongodb-starter-api** — Modular CRUD boilerplate with Express, MongoDB Mongoose, JWT auth, Redis cache, and Docker.
Tech Stack: Express, MongoDB Mongoose, JWT Auth, Redis Cache, Docker Compose

Directory Audit Note: Full source code, commit history, and test suites for all repositories are publicly available at github.com/mazkev and interactive web workstation at mazkev.vercel.app.

Interactive API Contracts, Microservices Topologies, Schema Proofs & Rate Limiters



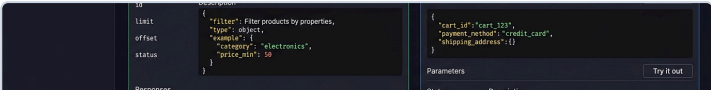
1. GoFinance Digital Wallet & Transfer API

BACKEND

Go • Echo • PostgreSQL • Redis • RabbitMQ • Docker

Digital wallet REST API simulation exploring ACID transactional transfers, Redis cache-aside ledger, and RabbitMQ broker.

github.com/mazkev/go-banking-core-system



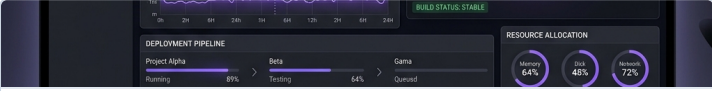
2. Swagger Go API Gateway Engine

BACKEND

Go 1.26 • Gin • GORM • PostgreSQL • Swagger OpenAPI

Production API gateway with interactive Swagger OpenAPI contract documentation, connection pooling, and JWT authorization.

github.com/mazkev/go-ecommerce-gateway-engine



3. Java Spring Commerce Platform

BACKEND

Java 17 • Spring Boot 3.3 • Vue 3 • PostgreSQL

Enterprise commerce and warehousing platform with Java 17, Spring Boot 3.3, Vue 3 Pinia, OpenPDF invoicing, and PostgreSQL.

github.com/mazkev/java-spring-commerce-platform



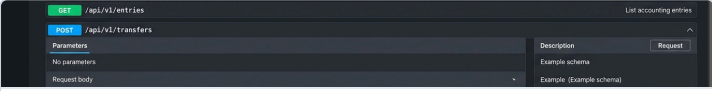
4. Go Clean Architecture Engine

BACKEND

Go • Gin • Clean Architecture • Docker • Unit Tests

Modular domain-driven design decoupling business logic, usecases, and repository data stores for maximum testability.

github.com/mazkev/go-clean-arch



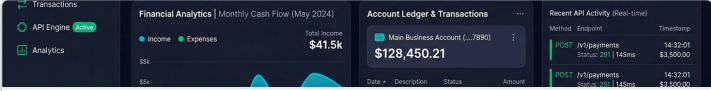
5. Digital Wallet Swagger UI & Transfer Ledger

BACKEND

Go • Echo • Swagger UI • Bcrypt PIN • Audit Logs

Interactive API testing suite verifying balance inquiries, atomic debit/credit transactions, and audit ledger entries in a simulated wallet environment.

github.com/mazkev/go-banking-core-system



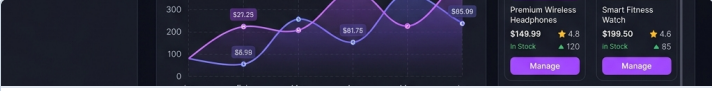
6. Microservices Concurrency Lab

BACKEND

Go • gRPC • Protocol Buffers • RabbitMQ • Redis

Binary inter-service communication pipeline with worker pool concurrency and decoupled background message queues.

github.com/mazkev/go-distributed-microservices-lab



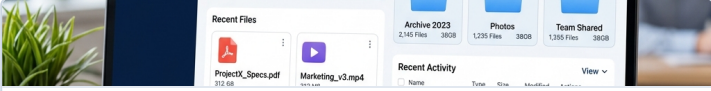
7. Bun Hono Ultra-Fast REST Engine

BACKEND

Bun • Hono v4 • Drizzle ORM • TypeScript • WebSocket

Sub-millisecond REST API engine running on Bun runtime with Drizzle ORM, live WebSocket chat, and coupon discount logic.

github.com/mazkev/hono-ecommerce-engine



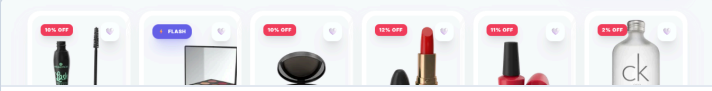
8. AI API Manager & Rate Limiter

BACKEND

Node.js • Express • Redis • Token Quotas • React Console

Reverse proxy API gateway with API key authentication, distributed rate limiting, token quota tracking, and latency analytics.

github.com/mazkev/AI-api-manager



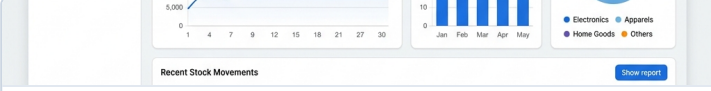
9. Midtrans Payment Gateway API

BACKEND

Node.js • Express v5 • Prisma ORM • Midtrans • PDFKit

Payment processing backend with Midtrans webhook verification, automated digital PDF invoice rendering, and email notifications.

github.com/mazkev/express-prisma-payment-api



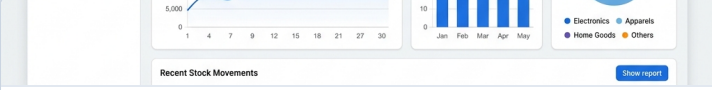
10. Spring Boot Enterprise Platform

BACKEND

Java 17 • Spring Boot 3.3 • JWT • Bucket4j • MongoDB

Enterprise platform with Spring Security JWT, AOP audit logging, async event-driven mailers, Bucket4j rate limiting, and Docker.

github.com/mazkev/spring-boot-enterprise-platform



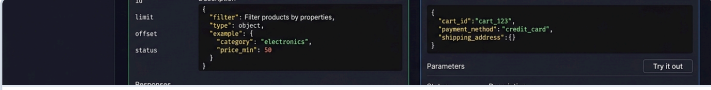
11. Spring Boot Book Catalog API

BACKEND

Java 17 • Spring Boot 3.3 • MongoDB • OpenAPI 3.0

Enterprise book catalog service with Spring Data MongoDB, automated OpenAPI documentation, and containerized deployment.

github.com/mazkev/spring-boot-book-manager-api



12. Go Enterprise REST API Boilerplate

BACKEND

Go • Gin • GORM • Redis Cache • Uber Zap • Swagger UI

Enterprise Go REST API boilerplate featuring Gin router, GORM ORM, Redis cache-aside, Uber Zap structured logging, and Swagger UI documentation.

github.com/mazkev/go-rest-api-enterprise

Interactive Demonstration & Source Code Audit: Live deployments, interactive case studies, architectural documentation, and full source code are accessible at mazkev.vercel.app and github.com/mazkev.